Home Assignment II

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Problem 1

See attached PDF file for the hand-written solution, as requested.

Problem 2: Relationship between paid parental leave and the gender pay gap

Problem 2.1: Theoretical background of the study

The task is to estimate the relationship between paid parental leave and the gender pay gap. The initial dataset contains 21 countries with variables measured in 2013, 2018, and 2023.
To find an unbiased ceteris paribus effect of paid parental leave on the gender pay gap, we have to think about the expected effect and necessary controls:

- *Paid Parental Leave (weeks)*: positive effect;
More paid parental leave weeks effectively implies that the person is unemployed for a longer time period which results in a more significant career break. This might make companies reluctant to hire the individual or hire them for a lower wage (due to lower experience vis-a-vis an individual who did not miss out as much). The effect should be diminishing as each additional week will not necessarily lead to linear skill eradication. For instance the cases with 2 and 3 years of leave can be treated similarly and considered as "people who stopped working for a long time".
- *Proportion of women in high-paying sectors*: negative effect;
One of the reasons for the difference in payouts between men and women is the fact that men tend to contribute to the more high-paying sectors in the economy. Therefore, if the share of women becomes higher in sectors, such as IT, finance, and law, (along with others), the pay gap should shrink. The effect should be roughly linear: with 1 pp increase in female proportion of specialists in high-paying sectors, the pay gap should decrease by x pp.
- *Governmental childcare policy*: negative effect;
If there is a childcare policy available, there is more chances the person will remain employed by outsourcing caring for a child to the governmental agency. This will prevent career break and promote smoother salary growth which should lead to lowering the gender pay gap. The effect should also be linear as the proportion of women otherwise not working remain employed and are subject to career growth, with no scaling or diminishing effects expected there.
- *Job-protected leave*: ambiguous effect;
On one hand, if the person has a job secured for the time of childcare, they do not face a career break, thus continue from the same point when returning to work. On the other hand, they might remain stuck at the same level of work, since they lost some skills while not working, so they will have to rebound and then start their career growth again. It will be difficult to change jobs and thus grow as the break from work will discourage companies to hire the individual right after a break. The effect should be diminishing similarly to paid leave.

Problem 2.2: Downloading the data

We downloaded the data from the Eurostat Database and the Excel file that was provided with the home assignment.
Here is the preview of the data:

```
Rows: 63
Columns: 9
$ country            <chr> "Austria", "Austria", "Austria", "Belgium", "Belgium", "Belgi~
$ year               <int> 2013, 2018, 2023, 2013, 2018, 2023, 2013, 2018, 2023, 2013, 2~
$ jobprotected_leave <dbl> 104.35714, 104.35714, 95.66071, 31.39286, 31.39286, 31.39286,~
$ paid_leave         <dbl> 122.446429, 113.571429, 113.571429, 17.392857, 17.392857, 17.~
$ child_forml_care   <dbl> 83.3, 80.0, 75.9, 54.3, 45.6, 43.7, 97.6, 90.9, 95.6, 35.1, 3~
$ gdp_per_capita     <dbl> 34310.7, 38414.6, 46171.3, 31661.5, 35574.9, 45291.9, 22410.8~
$ gender_pay_gap     <dbl> 22.3, 20.4, 18.3, 7.5, 5.8, 0.7, 22.3, 20.1, 18.0, 16.5, 14.6~
$ ict_spec_sex       <dbl> 14.2, 18.4, 19.5, 15.5, 16.8, 19.4, 10.6, 9.7, 12.4, 18.9, 20~
$ gender_empl_gap    <dbl> 9.1, 9.0, 7.8, 10.2, 8.4, 7.6, 17.2, 15.2, 13.9, 6.3, 7.0, 5.~
```

Problem 2.3: Downloading the data

- **Job-protected leave (weeks)**:
  Number of weeks during which parents can interrupt work for childcare while their original job position is legally guaranteed.
- **Paid leave (weeks)**:
  Number of weeks of parental leave that are financially compensated, capturing overall generosity of income support during leave.
- **% of children up to 3 years who do not receive childcare**:
  Share of young children (0–3) not enrolled in formal childcare or early education, used as an inverse proxy for childcare availability and usage.
- **GDP per capita EU PPS standard**:
  GDP per person expressed in Purchasing Power Standard, ensuring comparability in real income levels across countries; used as a control for economic development.
- **Unadjusted gender pay gap**:
  Difference in average gross hourly earnings between men and women, expressed as a percentage of male earnings; reflects overall gender disparity in wages without controlling for characteristics.
- **Employed ICT female specialists**:
  Share of ICT specialists who are women; indicates female representation in a high-paying, high-skill sector.
- **Gender employment gap**:
  Difference in employment rates between men and women (men minus women), capturing gender differences in labour market participation.

To discuss the units of the imported variables, let us look at their distributions. Based on skewness and dispersion, we shall be able to decide.

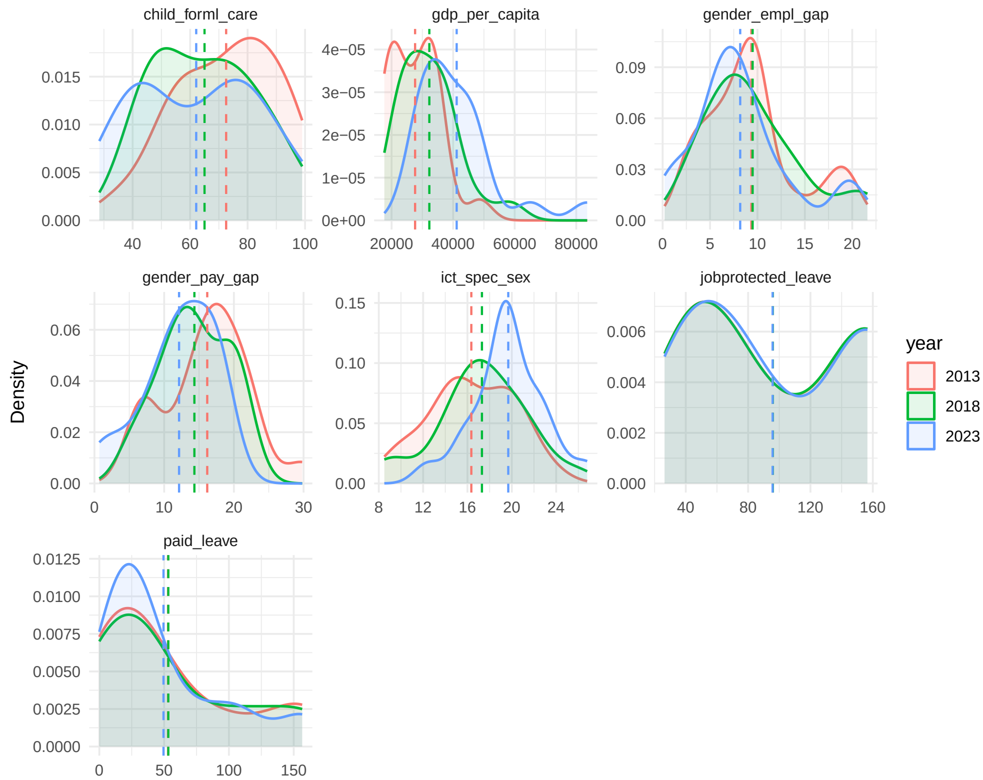

Figure 1: Density of all variables

After examining all distributions, we believe the one right-skewed with a big dispersion is `gdp_per_capita`, so we are going to log it.

Next we will consider scatterplots with the response variable, `gender_pay_gap`.

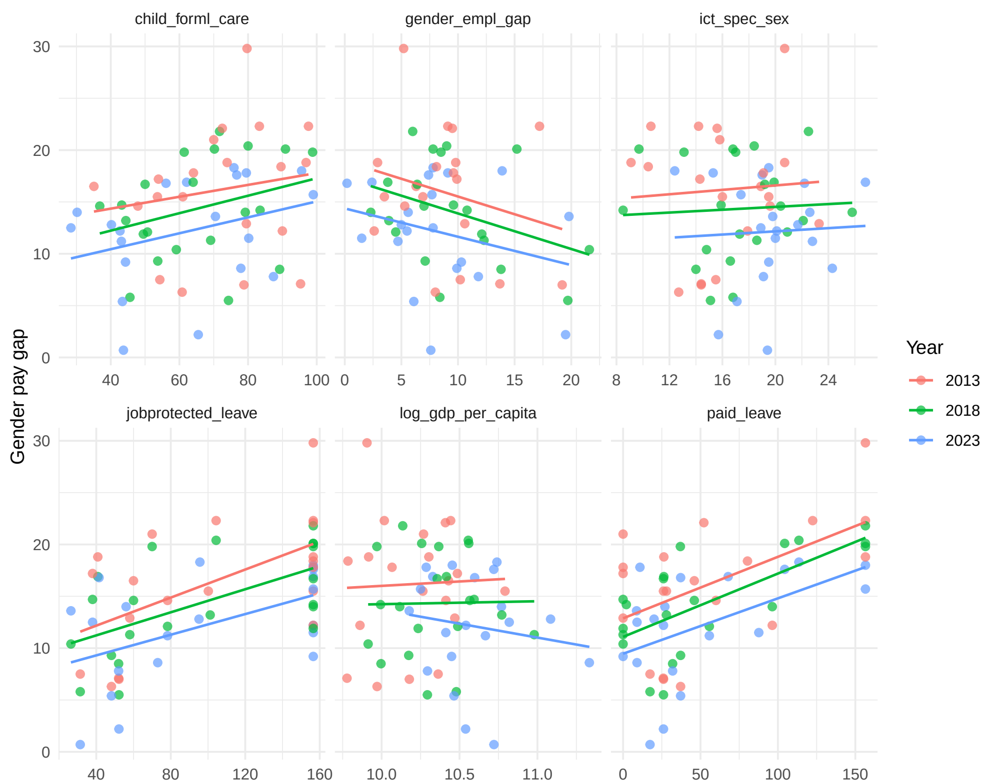

Figure 2: Scatterplots of all variables with gender_pay_gap, separated by years

Figure 2 suggests that there should be no structural breaks in the data, the slopes remain the same while the intercepts vary. This could be caused by the time trend in the variables that should be eliminated by FE, FD estimation.

We will also examine the box and jitter plots of the variables to see whether there are any outliers in the data.

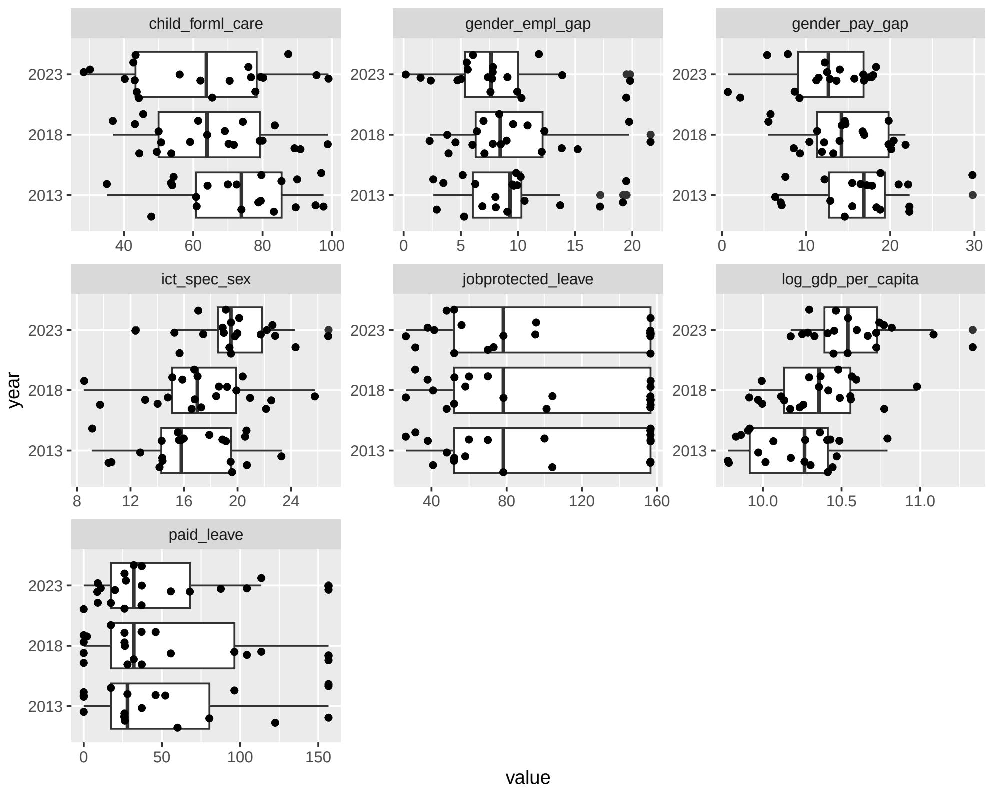

Figure 3: Box and jitter plots of all variables, separated by years

Box and jitter plots do not suggest the presence of any outliers that are worth deleting for the sample `n=20` over `T=3`. At the same time, we will spot that the log transformation helped standardize `gdp_per_capita` as it does not have a heavily-skewed distribution now.

Problem 2.4-2.5: First model and summary table

To report summary statistics, we shall add first differences and time demeaned variables to the dataset. This will allow not only see the distributions of original variables, but also consider the variables that will be used in the panel methods.

This is necessary because we need to check how much of variability remains from the original variables to expect the size of the SEs when estimating the models.

We will also delete *United Kingdom* from the observations since it has a lot of NA values for the year `2023` and does not have any values for `gender_empl_gap`.

Table 1: Columns with at least one NA for United Kingdom

child_forml_care	gender_pay_gap	ict_spec_sex	gender_empl_gap	log_gdp_per_capita
70.0	21.0	15.8	NA	10.26927
61.4	19.8	17.0	NA	10.36677
NA	NA	NA	NA	NA

Table 2: Summary statistcs (original, FD, time-demeaned)

Name	Min	QRT1.25.	Median	Mean	QRT3.75.	Max	NAs	SD
jobprotected__leave	26.393	51.036	86.723	97.084	156.536	156.536	0	52.281
paid__leave	0	17.393	30	53.291	89.835	156.536	0	52.070
child__forml__care	28.400	49.875	69.650	66.660	80.050	99	0	19.517
gender__pay__gap	0.700	10.800	14.200	14.012	17.800	29.800	1	5.551
ict__spec__sex	8.500	15.250	18.750	17.792	20.175	26.800	0	4.008
gender__empl__gap	0.200	5.575	8.050	9.008	10.650	21.600	0	5.003
log__gdp__per__capita	9.778	10.164	10.411	10.370	10.557	11.333	0	0.328
jobprotected__leave__cen	-5.798	0	0	0	0	10	0	1.983
paid__leave__cen	-59.119	-2.958	0	-0	2.714	49.245	0	13.896
child__forml__care__cen	-13.433	-3.833	-0.183	-0	3.567	13.800	0	6.001
gender__pay__gap__cen	-5.933	-1.250	-0.033	0	1.600	6.967	1	2.170
ict__spec__sex__cen	-4.100	-1.450	-0.217	-0	1.650	4.567	0	2.117
gender__empl__gap__cen	-2.133	-0.717	-0.067	0	0.675	1.767	0	0.947
log__gdp__per__capita__cen	-0.458	-0.136	-0.027	-0	0.165	0.406	0	0.176
jobprotected__leave__fd	-8.696	0	0	-0.051	0	15	20	3.013
paid__leave__fd	-88.679	0	0	-2.782	0	52.179	20	21.164
child__forml__care__fd	-26.400	-9.300	-6.350	-5.272	-1.875	11.400	20	6.925
gender__pay__gap__fd	-8	-2.700	-2.100	-1.800	-0.650	3.600	21	2.465
ict__spec__sex__fd	-5.800	0.300	1.700	1.653	3.925	7.900	20	3.066
gender__empl__gap__fd	-3.600	-1.700	-0.800	-0.587	0.250	2.700	20	1.447
log__gdp__per__capita__fd	-0.021	0.132	0.200	0.199	0.248	0.510	20	0.095

First model (light) that we propose will be the following:

$$gender\_\_pay\_\_gap_{2023,i} = \beta_0 + \beta_1 \cdot jobprotected\_\_leave_{2023,i} + \beta_2 \cdot paid\_\_leave_{2023,i} + u_t$$

We shall see the effect of `paid_leave` on `gender_pay_gap`, controlling for `jobprotected_leave` on the cross-section of the 20 countries in the dataset.

Table 3: First model results (cross-section, 2023)

	Dependent variable:
	Gender pay gap
jobprotected_leave	0.033
	(0.023)
	p = 0.157
paid_leave	0.035
	(0.025)
	p = 0.177
Constant	7.149
	(2.138)
	p = 0.004***
Observations	20
R^2	0.335
Adjusted R^2	0.257
Note:	*p<0.1; **p<0.05; ***p<0.01

Both of the explanatory variables appear to be statistically insignificant. That might be because of the small sample size `n=20` and thus high SEs.

Problem 2.6: First model and estimates for all years

Given that we have can exploit the time dimension of our data, we will estimate the same model accounting for a possible time trend in our variables. For the latter purpose, we will add year dummies to our equation.

In general, panel data should allow us to work with constant effects that influence each observation differently, explain the response variable and which are difficult to estimate. In the cross-sectional case, it is almost impossible to work with unchanging effects if you cannot proxy, thus they might create unsolvable OVB.

Since the scatterplots did not show any hint of changing slopes for the `jobprotected_leave` and `paid_leave`, we will not include their interactions with the dummies in the model.

$$gender\_pay\_gap_{t,i} = \beta_0 + \beta_1 \cdot jobprotected\_leave_{t,i} + beta_2 \cdot paid\_leave_{t,i} + \beta_3 t_{2018} + \beta_4 t_{2023} + u_t$$

To compare, we report both models, with and without year dummies. From the results (with dummies) in Table 4, we observe that both `jobprotected_leave` and `paid_leave` have a positive significant effect on `gender_pay_gap`. The former, with the one week increase, yields a 0.037 pp widening in gender pay gap while the latter only increases it by 0.041 pp.

The year dummies represent that gender pay gap shrinks every five years from 2013 to 2023 by roughly 1.5 pp. One of the dummies is insignificant which suggests that the first five-year impact is possibly due to data noise. We recognize that the trend can be explained with other factors that will be added to the model.

Overall, dummies allow us to account for time trend; however, do not change the coefficient of the models much as expected from the scatterplots (Figure 2), just “unbias” the intercept.

Table 4: First model results, all years (with and without year dummies)

	Dependent variable:	
	Gender pay gap	
	No year dummies	With year dummies
jobprotected_leave	0.036	0.037
	(0.013)	(0.013)
	$p = 0.008^{***}$	$p = 0.006^{***}$
paid_leave	0.043	0.041
	(0.013)	(0.013)
	$p = 0.002^{***}$	$p = 0.002^{***}$
factor(year)2018		−1.536
		(1.311)
		$p = 0.247$
factor(year)2023		−3.294
		(1.313)
		$p = 0.016^{**}$
Constant	8.096	9.806
	(1.191)	(1.404)
	$p = 0.000^{***}$	$p = 0.000^{***}$
Observations	59	59
R^2	0.436	0.495
Adjusted R^2	0.415	0.457
Note:		*p<0.1; **p<0.05; ***p<0.01

Problem 2.7: Main model - Fixed Effects (Gender Pay Gap ~ Paid Leave)

We propose the FE as the main model for finding the effect of *Paid Leave* on *Gender Pay Gap* in the following specification:

$$gender\_pay\_gap_{t,i} = \beta_0 + \beta_1 \cdot paid\_leave_{t,i} + \sum_j \beta_j x_{j,t,i} + \sum_t \beta_t year_t + u_t$$

We believe the best model from all ablations is FE with all controls in place, accounting for the time trend.

- *Fixed effects*:
  We select fixed effects as the model since we believe that there will be an unchanged variable a_i that will influence the reponse variables in all time periods, leading to serial correlation and violation of the exogeneity assumption.
  The example of a_i could be social institutions: treatment of women with relation to the labour market, in particular whether the labour of women and men costs the same, who has to stay with the kids as they are born, and should women go study STEM or do more people-oriented work, etc. Even though we have a cross-section of quite progressive European countries, there will still be a disparity between more egalitarian Nordic countries, Western Europe and more conservative Central and Eastern European countries. Furthermore, this disparity may not have the direction that we anticipate, as developing countries sometimes exhibit more egalitarianism. This may be because in

developed countries, individuals have more liberty to choose jobs that appeal to them more rather than choosing jobs necessary for prosperity.

- *Fixed effects*: It is important to account for time trend as this may lead to a spurious correlation problem. We can see it well on the example of `log_gdp_per_capita`. In the Table 5, the first model with individual effects, shows the impact of `log_gdp_per_capita` on the `gender_pay_gap`. On the other hand, we can see very well increasing trend of GDP and decreasing of Pay Gap which can lead to spurious correlation.

In the end, we decide to select the model with the most controls, to reduce the bias of the effect of

Table 5: Fixed effects model ablations

	Dependent variable:			
	Gender pay gap			
	FE baseline (2.6 task), ind effects	FE baseline	FE ict, gdp	FE all vars
child_forml_care	0.130			0.063
	(0.065)			(0.067)
	$p = 0.054^{*}$			$p = 0.359$
log_gdp_per_capita	−4.670		4.133	7.763
	(2.432)		(4.093)	(4.836)
	$p = 0.064^{*}$		$p = 0.320$	$p = 0.119$
paid_leave	0.029	0.034	0.035	0.035
	(0.020)	(0.018)	(0.018)	(0.018)
	$p = 0.155$	$p = 0.067^{*}$	$p = 0.055^{*}$	$p = 0.066^{*}$
jobprotected_leave	−0.056	−0.085		−0.239
	(0.143)	(0.125)		(0.147)
	$p = 0.698$	$p = 0.504$		$p = 0.114$
gender_empl_gap	0.185			−0.079
	(0.367)			(0.370)
	$p = 0.618$			$p = 0.834$
ict_spec_sex	0.069		0.298	0.299
	(0.178)		(0.168)	(0.182)
	$p = 0.702$		$p = 0.084^{*}$	$p = 0.110$
Observations	59	59	59	59
R^2	0.513	0.098	0.172	0.249
Adjusted R^2	0.144	-0.495	-0.412	-0.406
Note:				*p<0.1; **p<0.05; ***p<0.01

Problem 2.8: Assumptions disussion

Before the discussion of the assumptions, we will perform a short visual check to motivate our arguments for validity of our model.

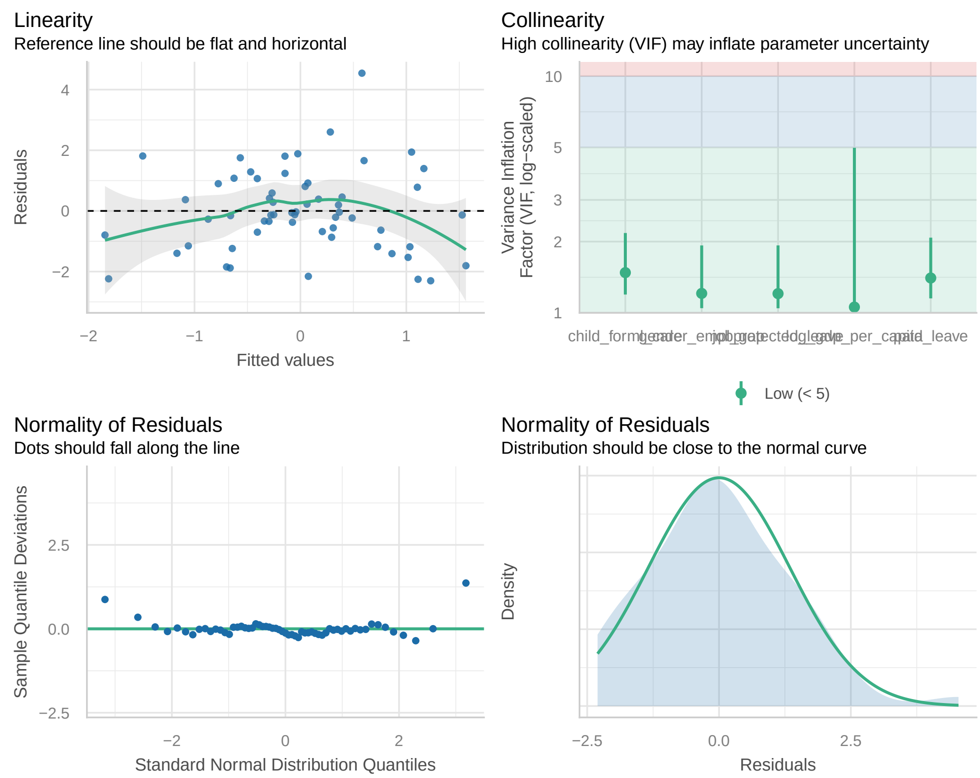

Figure 4: Visual check on the "FE all vars" model robustness

We observe that the model does not have collinearity problems, residuals also have a centered distribution around zero value, with no strong skew. At the same time, while there are linearity issues, there are also more extremely negative or positive residuals which might hint at functional misspecification or overall poor fit of the model.

- **Model specification:**

Since we have panel data and we argue for the existence of the unchanged effects specific to observations, we should consider FE and FDE. However, we also believe that sample `n=20` size is rather small and decreasing it further with the FDE is a bad idea, so we will stick to the FE. For presence of a_i will be conducted later via the Hausman test to compare the results of FE and RE as RE is more efficient in the absence of $E(a_i|X_i) \neq 0$.

- **Random sample:**

We believe the sample is random, in fact, we cover almost whole population as we have 20 countries in the dataset. More advanced as well as developing economies are included, thus there should be no selection bias due to factors in distrurbance that typically bias the estimators.

- **Variability of explanatory variables:**

Referring to Table 2, we observe proper variability in all variables.

- **Exogeneity:**

Referring to the argument made in section 2.7, we believe that a_i is part of the disturbance since, for example, social attidute to women on the labour market can relate to `gender_pay_gap` and possibly to the level of economic development (`log_gdp_per_capita`) as well as `gender_empl_gap`, `ict_spec_sex`, etc.

- **Homoscedasticity:** We shall conduct Breush-Pagan test to know whether the variance of errors is homoscedastic.

$p - value:\ 0.4872 \implies$ errors are homoscedastic.

- **Serial correlation:** We will conduct Wooldridge test for serial correlation[1].

$p - value = 0.00567 \implies$ serial correlation.

This will require us to report heteroscedasticity and autocorrelation robust standard errors.

Overall, we believe the model has a poor fit of the data, with the negative R^2 after we account for the time trends of the variables. The effect of `paid_leave` remains relatively stable, regardless of the changes in the model which means that the other explanatory variables have low correlation with `paid_leave` and they are also poor control for the `paid_leave` as all of them appear insignificant.

The negative adjusted R^2 should not be a major concern in this context. Adjusted R^2 measures the proportion of total variation in the dependent variable explained by the model, but our primary objective is not to maximize overall explanatory power. Instead, we focus on identifying the effect of the `paid_leave`on `gender pay gap`, specifically its direction, magnitude, and statistical significance. Therefore, even if the variable contributes little to the model's total explained variance, it remains central to our research question as long as its estimated coefficient is interpretable and statistically reliable.

Problem 2.9: RE vs FE

In the following section we are going to test for the existence of a_i. For that matter we are going to run RE specification and conduct a Hausman test.

$p - value = 0.04555 \implies$ one of the models is inconsistent due to the a_i which is RE because FE controls for it.

Otherwise, we have 2 specifications: the one that we received from a previous section with all available variables and the one where we drop `child_form_care` and `gender_empl_gap`.

We will argue that the model with less variables is better. `child_form_care` and `gender_empl_gap` remain insignificant even after the treatment with heteroscedasticity and autocorrelated standard errors which we specify with small-sample correction and assuming independence between the observations.

[1] not Durbin-Watson test, as it does not work for panel linear regressions

Table 6: FE and RE model ablations, HAC SEs

	Dependent variable:		
	Gender pay gap		
	FE main	FE all vars	
child_forml_care		0.063	
		(0.050)	
		$p = 0.206$	
log_gdp_per_capita	7.833	7.763	−5.628
	(3.629)	(3.754)	(1.774)
	$p = 0.031^{**}$	$p = 0.039^{**}$	$p = 0.002^{***}$
paid_leave	0.040	0.035	0.038
	(0.021)	(0.021)	(0.017)
	$p = 0.064^{*}$	$p = 0.086^{*}$	$p = 0.026^{**}$
jobprotected_leave	−0.213	−0.239	0.030
	(0.071)	(0.089)	(0.020)
	$p = 0.003^{***}$	$p = 0.008^{***}$	$p = 0.138$
gender_empl_gap		−0.079	
		(0.298)	
		$p = 0.792$	
ict_spec_sex	0.345	0.299	0.040
	(0.144)	(0.143)	(0.122)
	$p = 0.017^{**}$	$p = 0.037^{**}$	$p = 0.745$
Constant			66.741
			(17.728)
			$p = 0.0002^{***}$
Observations	59	59	59
R^2	0.226	0.249	0.381
Adjusted R^2	-0.360	-0.406	0.336
Note:			*p<0.1; **p<0.05; ***p<0.01

Overall, from 2 specifications of FE and RE, we believe that the one with less variables, `FE_main` is better since it has significant controls and unbiases the effect of the `paid_leave` properly while the other FE specification with all variables simply increases multicollinearity, thus reducing statistical significance of the relevant variables, without having meaningful effect on the response.

Problem 2.10

- **Estimates:**

Lastly we believe given our data, specification `gender_pay_gap ~ log_gdp_per_capita + paid_leave + jobprotected_leave + ict_spec_sex` is the best. It is FE estimator with within group heteroscedasticity and autocorrelation robust SEs.

The model finds that increase in one week of Paid Parental Leave leads to 0.04 pp increase in the Gender Pay Gap. It aligns with our intuition since the payment, that is typically received by women, incentivizes them to stay out of work more and have a larger career break which leads to the period of recovery and lower salary growth.

Job Protected Leave has a negative effect. Our reasoning was ambiguous as to whether the effect should be posisitive or negative. It is likely because staying with the job you have smoothes the career recovery phase after getting back to work.

Proportion of female specialists in high-paying jobs increases the Gender Pay Gap which is an unexpected result. Since the gap has the following formula $\frac{w_{men}-w_{women}}{w_{men}} \cdot 100$, `ict_spec_sex` might have a correlation with the omitted variable, which biases upwards its coefficient significantly. Also, even though women are in higher-paying sector, they may have junior / support / part-time roles while men in senior / bonus-heavy / overtime roles which may lead to increase in the gap.
Lastly, `log_gdp_per_capita` has a positive effect. The reason could be due to richer societies typically being more unequal, which supports our discussion of the direction of the effect in the fixed effects section 2.7. And since the income distribution is highly right-skewed, the men can dominate it and increase the gap.

- **What we would do differently?**
- *Data*: we would import more data, have more variables, and have it more granular (sector data, all EU countries, richer time dimension). This could lead to more consistent estimates, thus better validity of all tests and statistics.
- *Models*: we would estimate FD with richer data to see whether is is more significant or has a higher R^2.
- *Specifications*: we would consider different functional specifications and interaction effects theoretically to build a better, more robust model.